

MEAM 5200-Lab1: Forward Kinematics

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1 Introduction

Forward Kinematics (FK) is a robotic technique that uses joint parameters and link lengths to calculate the position of the end effector of the robot. The purpose of this lab was to implement the forward kinematics of the Franka Emika Panda robot arm and evaluate the implementation using both simulation and hardware testing. The simulation utilized the FK method developed in Python and the result will then be compared with measurements from the physical robot arm under identical configurations. Three different configurations were tested to evaluate the accuracy and consistency of the implementation.

2 Method

2.1 Coordinate Frame assignment and DH parameters

The forward kinematics of the Franka Emika Panda arm were computed using the standard Denavit-Hartenberg (DH) convention.

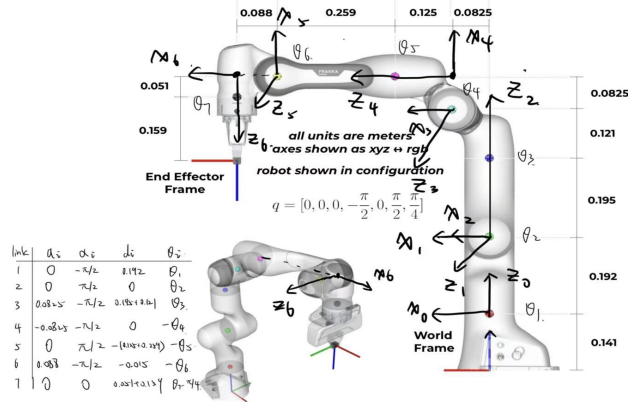


Figure 1: Coordinate Assignment

The DH parameters were derived directly from the robot dimensions shown in the schematic. The end-effector frame is rotated by a constant 45 degrees relative to the final joint frame, which was incorporated as a constant angular offset in joint 7. Table 1 lists the link parameters and joint angle mappings used to construct the homogeneous transformation matrices between consecutive frames.

Table 1: Denavit–Hartenberg parameters for the Franka Emika Panda arm.

i	a_i (m)	α_i (rad)	d_i (m)	θ_i
1	0	$-\frac{\pi}{2}$	0.192	q_1
2	0	$\frac{\pi}{2}$	0	q_2
3	0.0825	$-\frac{\pi}{2}$	0.316	q_3
4	-0.0825	$-\frac{\pi}{2}$	0	$-q_4$
5	0	$\frac{\pi}{2}$	-0.384	$-q_5$
6	0.088	$-\frac{\pi}{2}$	0	$-q_6$
7	0	0	0.210	$q_7 - \frac{\pi}{4}$

2.2 Forward Kinematics Formulation

The homogeneous transformation between consecutive frames is computed using the standard Denavit–Hartenberg formulation:

$$A_i = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\alpha_i)$$

$$A_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The base frame is offset from the world frame by a vertical translation:

$$T_{\text{world} \rightarrow 0} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.141 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Each link transformation is computed as:

$$T_{i-1,i} = A_i(\theta_i, d_i, a_i, \alpha_i)$$

$$T_{01} = A_1, \quad T_{12} = A_2, \quad T_{23} = A_3, \quad T_{34} = A_4, \quad T_{45} = A_5, \quad T_{56} = A_6, \quad T_{67} = A_7$$

2.3 Code implementation

Boyan Zhang’s code was used for experiments. The `calculateFK.py` file implements forward kinematics for a 7-DOF robotic arm using the Denavit-Hartenberg (DH) convention. The `FK` class initializes DH parameters (a, α, d) for all seven joints in its constructor, along with a world-to-base transformation matrix. A static helper method `_A` computes individual 4×4 homogeneous transformation matrices from DH parameters using standard trigonometric relationships. The main `forward` method accepts a 7-element joint angle vector, applies joint-specific transformations, sequentially multiplies transformation matrices to propagate from the base frame through each joint, extracts joint center positions from the translation components of each transformation matrix (with additional offsets for specific joints), and returns both an 8×3 matrix of joint positions in world coordinates and the 4×4 end-effector transformation matrix `T0e`.

3 Evaluation

To test the validation of the forward kinematics implementation, 3 joint configurations were selected and tested in both simulation and hardware. For each configuration, the robot was first moved in simulation using `demo.py`, and then executed on the physical robot via the `franka_interface`. The same joint vector was input into the `calculateFK` function to compute the predicted end-effector pose. The computed pose was then compared with the simulated and physical robot configurations.

3.1 Zero configuration

For zero configuration, all the joint angles are assigned to be 0. The configuration vector therefore has the form:

$$q_0 = [0, 0, 0, 0, 0, 0, 0]$$

The forward kinematics function produced the following homogeneous transformation matrix:

$$T_e^0 = \begin{bmatrix} 0.7071 & 0.7071 & 0 & 0.088 \\ 0.7071 & -0.7071 & 0 & 0 \\ 0 & 0 & -1 & 0.823 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The rotational matrix $R = \begin{bmatrix} 0.7071 & 0.7071 & 0 \\ 0.7071 & -0.7071 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is orthonormal which confirm this is a valid rigid body rotation, and the end effector position in the world frame is:

$$p = [0.088, 0, 0.823] \text{ m}$$

The end effector position indicates that the robot is primarily extended in the vertical direction, which is consistent with the expected zero configuration geometry. The small x offset corresponds to the wrist link dimension. Both the position and orientation components of the T_e^0 are consistent with the expected physical configuration of the robot in the zero pose.

3.2 Configuration 1

Configuration 1 has the configuration vector:

$$q_1 = [0, 0, 0, -1.57, 0, 1.57, 0.785]$$

The forward kinematics produced the homogeneous transformation with the end effector position

$$p = [0.554, 0, 0.522] m$$

Compared to the zero configuration, the end effector moved significantly in the x direction from 0.088m to 0.554m while lowered slightly in the z direction from 0.823 to 0.522 which indicates the arm extends forward and bends downward which is consistent with the physical configuration as show in Figure 3. The measured end effector position is $[0.472, 0, 0.515] m$ with largest error of 14.8 % which can be considered close to the code calculated position.

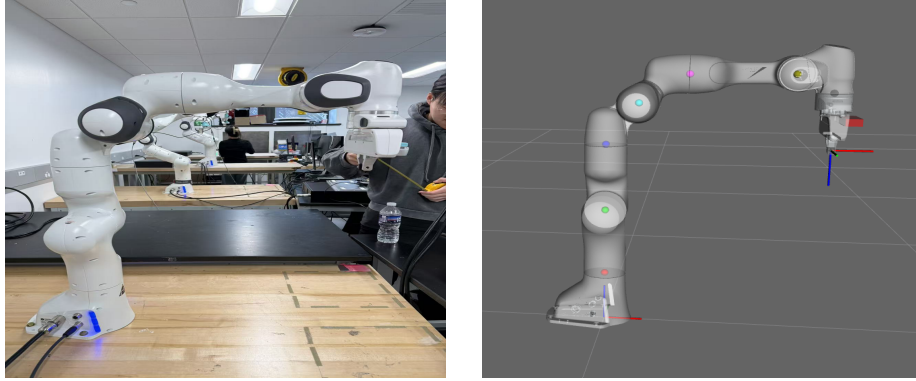


Figure 2: Configuration 1: Physical hardware (left) and Simulation (right).

The simulation result also matches with the physical configuration as can be seen in Figure 2.

3.3 Configuration 2

Configuration 2 has the configuration vector:

$$q_2 = [0, -1, 0, -1, 0, 1, 1]$$

The forward kinematics produced the homogeneous transformation with the end effector position

$$p = [-0.0796, 0, 0.918] m$$

The end effector position value of -0.0796m in x direction and 0.918m in z direction indicates that the arm bends upward and slightly backward, which is consistent with the physical configuration as show in Figure 4. The measured end effector position is $[-0.094, 0, 0.905] m$ with largest error of 18% which can be considered close to the code calculated position.

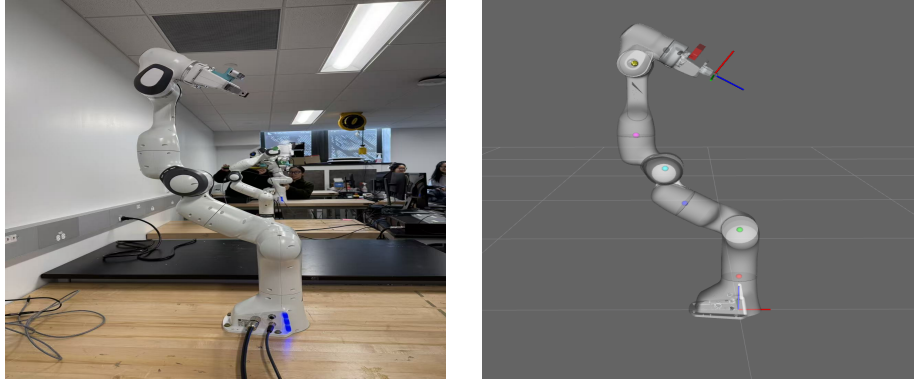


Figure 3: Configuration 2: Physical hardware (left) and Simulation (right).

The simulation result also matches with the physical configuration as can be seen in Figure 3.

3.4 Configuration 3

Configuration 3 has configuration vector:

$$q_3 = [0, -1, -1, -2, 0, 2, 1]$$

The forward kinematics produced the homogeneous transformation with the end effector position

$$p = [0.1438, -0.466, 0.596] m$$

The minus value in the y direction indicates the robot arm rotates toward the right, while the positive values in x and z directions indicates the arm extend forward and upward in those directions. Which the calculated position prediction matches with the physical configuration as show in Figure 5. The measured end effector positions is $[0.153, -0.508, 0.584] m$ with largest error of 9% which can be considered close to the code calculated position.

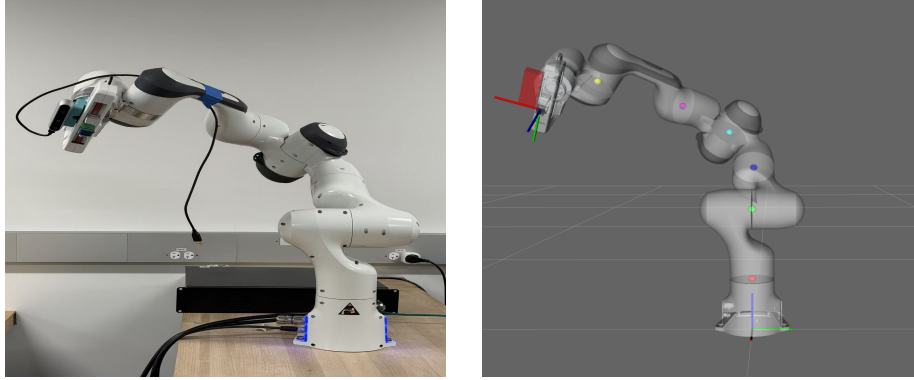


Figure 4: Configuration 3: Physical hardware (left) and Simulation (right).

The simulation result also matches with the physical configuration as can be seen in Figure 4.

3.5 Reachable workspace

The workspace visualization strategy employs a Monte Carlo sampling approach to approximate the reachable workspace of the 7-DOF Panda robotic arm. The implementation assumes that uniformly sampling joint configurations within the specified joint limits and mapping them to end-effector positions via forward kinematics provides a representative visualization of the workspace. Specifically, the algorithm generates 10,000 random joint angle configurations by sampling each of the seven joint angles independently from a uniform distribution bounded by the provided lower and upper limits. For each sampled configuration, the forward kinematics solver computes the corresponding end-effector position in the world frame by extracting the translation components (x, y, z) from the homogeneous transformation matrix $T0e$. The resulting end-effector positions are then visualized as a 3D scatter plot, where point density indicates regions of higher reachability. Key assumptions include: (1) uniform random sampling adequately captures the workspace geometry without systematic bias, (2) 10,000 samples provide sufficient point density for meaningful visualization, (3) the provided joint limits accurately represent the physical constraints of the robot, (4) the forward kinematics implementation correctly maps joint space to Cartesian space, and (5) the discrete point cloud representation is sufficient for understanding the workspace boundaries and density distribution, even though it does not guarantee complete coverage of all reachable points.

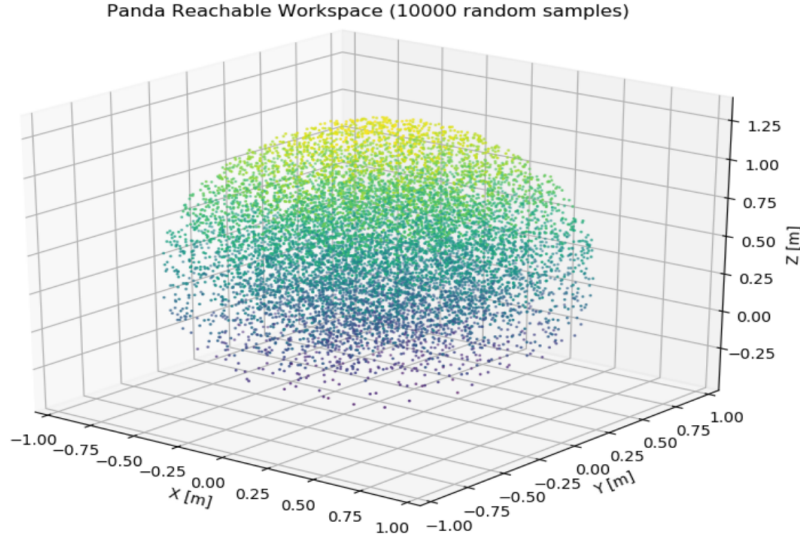


Figure 5: Franka Emika Panda Reachable Workspace

4 Analysis

4.1 Result analysis

Overall, the forward kinematics results were consistent with geometric expectations. The zero configuration produced an upright posture, and the end effector position matched the calculated result. For configuration 1 2 and 3, the predicted motion trends were also physically reasonable. All the tested configurations show agreement in simulation and actual hardware measurement, which suggest that the kinematic chain and transformation order were implemented correctly.

Although the qualitative agreement between simulation and hardware measurement was strong, quantitative discrepancies of approximately 9-18% were observed. These differences are likely due to measurement error caused by visual difference, misalignment of reference frame, or slight mismatch of the calibration of the physical robot.

4.2 Theoretical and simulated robot workspace

The workspace generated using forward kinematics code represents an ideal geometric workspace based on joint limits and rigid-link assumptions. How-

ever, in simulation, additional constraints can reduce or reshape the reachable region, the constraints include self-collision, environment collision, controller safety limit, etc. Therefore, some points predicted as reachable by theoretical forward kinematics may not be reachable in the simulation. To predict these differences, the collision geometry and joint limits need to be considered in the code.

4.3 Simulation VS Hardware

Simulation result closely matched the forward kinematics prediction because both rely on the same idealized kinematic model. In contrast, hardware testing introduces real-world effects such as mechanical backlash, structural compliance, and measurement error, which are not captured in a purely geometrical forward kinematics approach.